1d Coordinate Geometry

Ankit

MOP 2025

dont worry abt config issues and directing lengths properly for now, u have better things to do

§1 Warmup

Let ABC be a triangle. Compute BT/CT, where T is on BC and

1. T, A, and the circumcenter are collinear
2. TA is tangent to (ABC)
3. T, A, and the nine point center are collinear
4. T lies on the Euler line
5. TA is the radical axis of (AFE) and (ABC), where F lies on AB and E lies on AC (do this in two ways).
6. TA is tangent to (AFE), where F lies on AB and E lies on AC (do this in two ways).
7. $\angle BAT = \angle CAD$, where D is the A-intouch point
8. $TA = TD$, where D is your favorite point (lets say the A-intouch point again)
9. AT passes through a point on the incircle which is twice as close to the B-intouch point as it is to the C-intouch point.

And now a couple with T on the circumcircle:

10. T is collinear with A and your favorite point from the previous problems
11. T is on minor arc AC with $AT/CT = 2$
12. T lies on the A-midline

§2 Philosophy

Usually, a point in a geometry problem is on a line or a circle or something, which are 1d objects. So you should try to get really good at computing coordinates on lines and circles. Here are a couple examples of coordinates for a point P on a line ℓ; the first four are linear:

1. The x coordinate of P.
2. Length QP for a fixed point Q on ℓ
3. The area of $[ABP]$ for any points A, B.

4. The quantity $\text{pow}(P, \omega_1) - \text{pow}(P, \omega_2)$ for any circles ω_1, ω_2.

5. QP/RP for points Q, R on the line.

6. $\frac{af(P)+b}{cf(P)+d}$, where $f(P)$ is any other coordinate for P.

The situation for circles is a bit more dire; 5 (and 6) from above are sorta your only coordinates. But that's actually plenty. Also angles are sorta coordinates too. I'd tell you that $x + yi$ is a coordinate on a circle but I'm not sure you'd take me seriously.

§3 Degenerate Cubics

Collinearity is really nice on cubics. This includes degenerate cubics (e.g. three lines, a circle and a line). You probably know Menelaus, but here are some things you might not know:

Theorem 1 (Ratio Lemma)**.** *Let ω be a circle through B and C, and let a line meet ω at points A, D and BC at E. Then,*

$$\frac{BE}{CE} = \frac{BA}{CA} \cdot \frac{BD}{CD}.$$

Theorem 2. *Let $ABCD$ be a cyclic quadrilateral, and let E and F be points on line BC. If $ADEF$ is cyclic, then $\frac{BA}{CA} \cdot \frac{BD}{CD} = \frac{BE}{CE} \cdot \frac{BF}{CF}$.*

Theorem 3 (Ptolemy)**.** *Let O be a fixed point and ℓ_1, ℓ_2, ℓ_3 be fixed lines through it. Let θ_1, θ_2, and θ_3 be the angles between ℓ_2 and ℓ_3, ℓ_3 and ℓ_1, and ℓ_1 and ℓ_2. Points P_1, P_2, P_3 are chosen on each of these lines. Then, $OP_1P_2P_3$ is a cyclic quadrilateral if and only if*

$$\sin(\theta_1)OP_1 + \sin(\theta_2)OP_2 + \sin(\theta_3)OP_3 = 0.$$

Theorem 4. *Same setup. P_1, P_2, and P_3 are collinear if and only if*

$$\frac{\sin(\theta_1)}{OP_1} + \frac{\sin(\theta_2)}{OP_2} + \frac{\sin(\theta_3)}{OP_3} = 0.$$

§4 Final words of wisdom before problems

1. conics are just lines but placed in the plane weird

2. sometimes circles are just $x^2 + y^2$ plus some linear stuff, but sometimes they're something deeper

3. if you want to compute sin(alpha)/sin(beta), sometimes you can just find a triangle with angles alpha and beta. this goes beyond lengthbashing, sometimes if you want to understand something you should just find it

4. draw in the reims theorem points

5. all lengths secretly carry their arguments with them, we just don't write it down. so if anybody ever writes something like "$AB + AC$" they're choosing an angle bisector of $\angle BAC$. if you think about this hard enough you can, e.g. eliminate the need for directed lengths and make sin and cos be injective with period 180 (and hence be applicable to directed angles)

6. sometimes geometry problems are pretty deep and u cant just think in terms of coordinates, so u should be able to realize when thats happening. usually what i do is i try finding pairs of similar quadrilaterals (see wisdom #3), then i try complex bashing, then i give up. but i haven't done geometry in 5 years so that may be bad advice.

§5 Real Problems

If you get comfortable computing, these are all "just do it" problems. Problems are sorted by length.

Problem 5.1. *Let ABC be a triangle with incenter I. Show that the orthocenter of AIB, the orthocenter of AIC, and the A-intouch point are collinear.*

Problem 5.2. *Let $ABCD$ be a trapezoid with $AB||CD$, and let M be the midpoint of AC. Show that AD is tangent to the circumcircle of ABM if and only if BC is tangent to the circumcircle of CDM.*

Problem 5.3. *Let ABC be a triangle with circumcircle Ω. X and Y are points on Ω such that XY meets AB and AC at D and E, respectively. Show that the midpoints of XY, BE, CD, and DE are concyclic.*

Problem 5.4. *Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E. Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.*

Problem 5.5. *Let $\triangle ABC$ be a triangle with circumcenter O. The perpendicular bisectors of the segments OA, OB and OC intersect the lines BC, CA and AB at D, E and F, respectively. Prove that D, E, F are collinear.*

Problem 5.6. *Let $ABCDE$ be a convex pentagon such that $\triangle ABE$, $\triangle BEC$, and $\triangle EDB$ are similar (with vertices in order). Lines BE and CD intersect at point T. Prove that line AT is tangent to the circumcircle of $\triangle ACD$.*

Problem 5.7. *Cevians AP and AQ of a triangle ABC are symmetric with respect to its bisector. Let X, Y be the projections of B to AP and AQ respectively, and N, M be the projections of C to AP and AQ respectively. Prove that XM and NY meet on BC.*

Problem 5.8. *Let $BCDEF$ be a cyclic pentagon. Suppose DE and DF meet BC at X and Y, and let the circumcircle of DXY meet the circumcircle of the pentagon at Z. Show that E, F, the reflection of Z across BC, and the reflection of D across the midpoint of BC are concyclic.*

Problem 5.9. *Let ABC be a triangle and let M and N denote the midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Let X be a point such that $\overline{AX}$ is tangent to the circumcircle of triangle ABC. Denote by ω_B the circle through M and B tangent to $\overline{MX}$, and by ω_C the circle through N and C tangent to $\overline{NX}$. Show that ω_B and ω_C intersect on line BC.*

Problem 5.10. *Let $\triangle ABC$ be a triangle. Points D, E, and F are placed on sides $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$ respectively such that $EF \parallel BC$. The line DE meets the circumcircle of $\triangle ADC$ again at $X \neq D$. Similarly, the line DF meets the circumcircle of $\triangle ADB$ again at $Y \neq D$. If D_1 is the reflection of D across the midpoint of $\overline{BC}$, prove that the four points D, D_1, X, and Y lie on a circle.*

§6 Circus

Not real problems but maybe enlightening? idk.

Problem 6.1. *Let ABC be a triangle, and let ω be the circle of points T with $BT : CT = \lambda$. Prove that the power from A to ω is $\frac{AC^2 \cdot \lambda^2 - AB^2}{\lambda^2 - 1}$.*

Note: The intended solution is like 0 lines, you should sort of just realize its true.

Problem 6.2. *Prove theorem 3.2 from theorem 3.1*

Note: This is actually an exercise in synthetic geometry.

Problem 6.3. *Let $BPQC$ be on a line in that order, and pick R on the line such that $RB \cdot RC = RP \cdot RQ$. Prove, in as many ways as you can think of, that $\frac{BR}{CR} = \frac{BP}{CP} \cdot \frac{BQ}{CQ}$.*

Problem 6.4. *Think very deeply about the following 1d objects:*

1. *The family of lines tangent to a conic (should behave like points on a conic by duality, so you can define a cross ratio on them, etc).*

2. *The family of conics passing through four fixed points (should behave like a line).*

Problem 6.5. *Let O be a fixed point and ℓ_1, ℓ_2, ℓ_3 be fixed lines through it. Let θ_1, θ_2, and θ_3 be the angles between ℓ_2 and ℓ_3, ℓ_3 and ℓ_1, and ℓ_1 and ℓ_2. Points P_i and Q_i are chosen on line ℓ_i for each i. Prove that $P_1P_2P_3Q_1Q_2Q_3$ lie on a conic if and only if*

$$\frac{\sin(\theta_1)}{OP_1} + \frac{\sin(\theta_1)}{OQ_1} + \frac{\sin(\theta_2)}{OP_2} + \frac{\sin(\theta_2)}{OQ_2} + \frac{\sin(\theta_3)}{OP_3} + \frac{\sin(\theta_3)}{OQ_3} = 0.$$

Note: Similar things exist for Ceva's and the Ratio Lemma, but this is the only one for which I know a solution nicer than "use DDIT." I'd bet you can unify all these by thinking really hard about the Picard group, but I don't know algebraic geometry. Also the theorem is less impressive for points lying on a cubic or higher degree curves anyways, its stops being an iff. This is because 1 and 2 are the solutions to $3n = \binom{2+n}{2}$.

Problem 6.6. *Suppose $AA'BB'CC'XY$ lie on a circle in that order. Prove that AA', BB', and CC' concur if and only if*

$$\begin{vmatrix} XA \cdot XA' & XA \cdot YA' + XA' \cdot YA & YA' \cdot YA' \\ XB \cdot XB' & XB \cdot YB' + XB' \cdot YB & YB' \cdot YB' \\ XB \cdot XB' & XB \cdot YB' + XB' \cdot YB & YB' \cdot YB' \end{vmatrix} = 0.$$

Problem 6.7. *Let $I = (1 : i : 0)$, $J = (1 : -i : 0)$ be the circle points, and let A, B, C be noncollinear points.*

1. *What is $(AB \cap IJ, AC \cap IJ; I, J)$?*

2. *What is $(A, I; B, C)_{(ABC)}$?*

3. *Where does AI meet the x-axis?*

4. *Where do I and J go under rotations? Reflections?*

5. *Find all polynomials P with real coefficients such that*

$$\frac{P(x)}{yz} + \frac{P(y)}{zx} + \frac{P(z)}{xy} = P(x-y) + P(y-z) + P(z-x)$$

holds for all nonzero real numbers x, y, z satisfying $2xyz = x + y + z$.